16 Gradient Descent: Optimization Problems (not just) on Graphs

This chapter covers

- Developing a randomized heuristic to find the minimum crossing number
- Introducing cost functions to show how the heuristic works
- Explaining gradient descent and implementing a generic version
- Discussing strengths and pitfalls of gradient descent
- Applying gradient descent to the graph embedding problem

If I mention a technique called *gradient descent*, does it ring a bell? You might not have heard of, or maybe you recognize the name but can't quite recall how it works? If so, that's fine. If, however, I ask you about machine learning, classification problems or neural networks, chances are that you know exactly what I'm talking about: and I bet these terms also sparkled your interest much more.